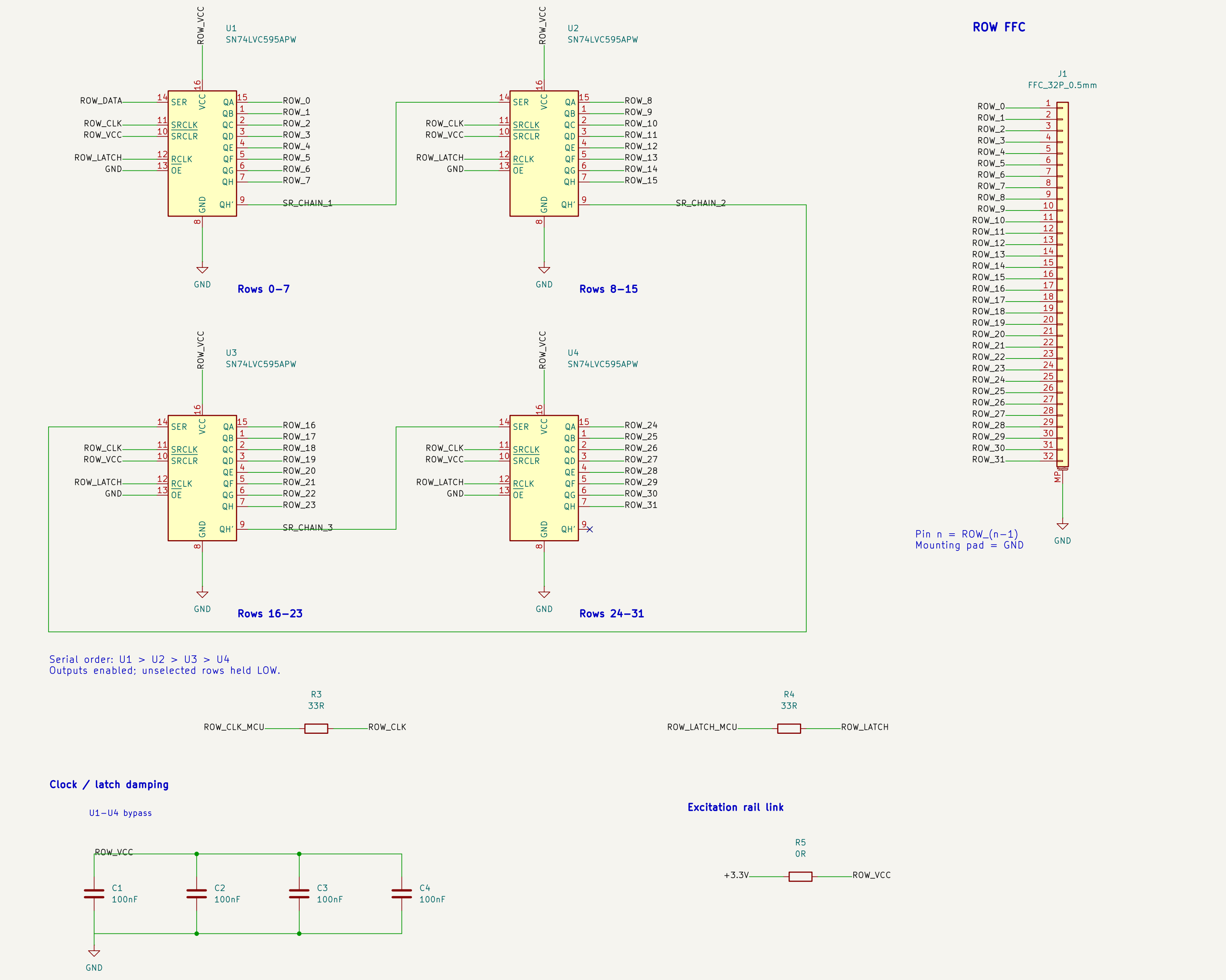
TAXELSCAN / REV 4

32 x 32 matrix > two sense banks > RP2354A internal 12-bit ADC > RS-485 harness

01 ROW EXCITATION

MCU serial stream > four latched drivers > 32-way row FFC



Pin n = ROW_(n−1)
Mounting pad = GND

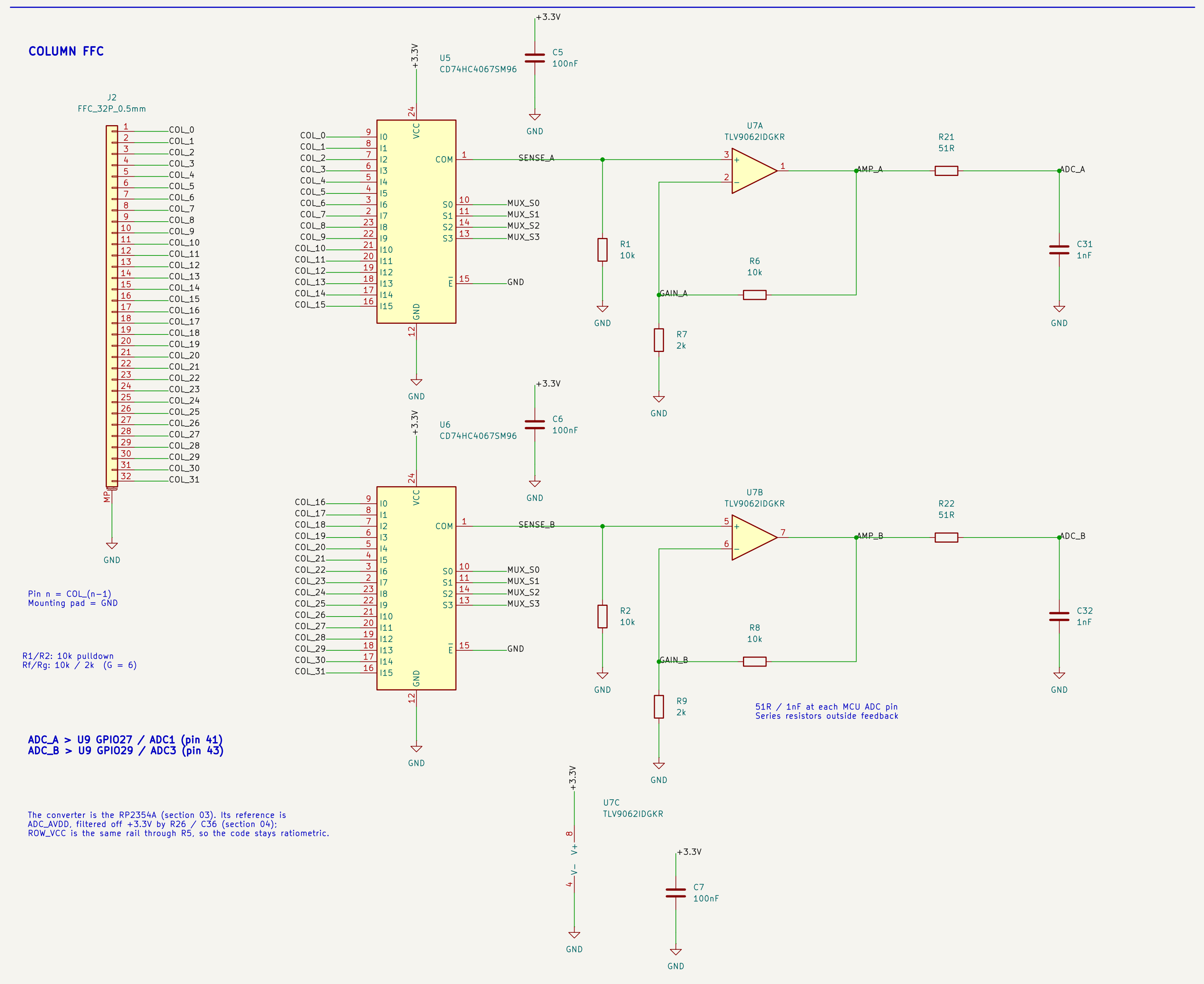
Serial order: U1 > U2 > U3 > U4
Outputs enabled; unselected rows held LOW.

Clock / latch damping

Excitation rail link

02 COLUMN SENSING

32 columns > two 16:1 muxes > gain of 6 > filtered inputs to the MCU ADC



Pin n = COL_(n−1)
Mounting pad = GND

R1/R2: 10k pulldown
Rf/Rg: 10k / 2k (G = 6)

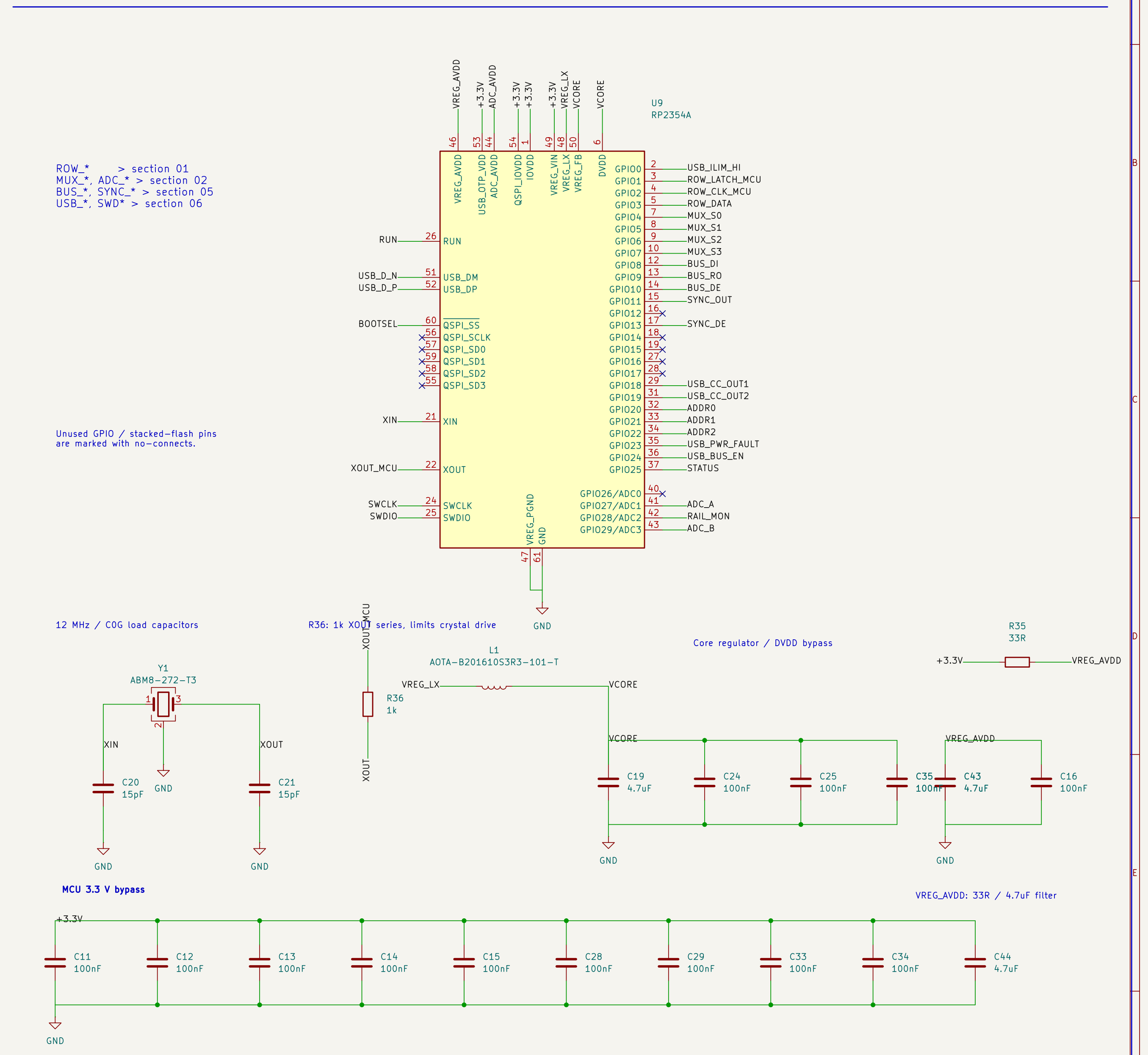
51R / 1nF at each MCU ADC pin
Series resistors outside feedback

ADC_A > U9 GPIO27 / ADC1 (pin 41)
ADC_B > U9 GPIO29 / ADC3 (pin 43)

The converter is the RP2354A (section 03). Its reference is
ADC_AVDD, filtered off +3.3V by R26 / C36 (section 04);
ROW_VCC is the same rail through R5, so the code stays ratiometric.

03 SCAN CONTROLLER

RP2354A / clock / core supply / local bypass



ROW_* > section 01
MUX_*, ADC_* > section 02
BUS_*, SYNC_* > section 05
USB_*, SWD* > section 06

Unused GPIO / stacked-flash pins
are marked with no-connects.

12 MHz / C0G load capacitors

R36: 1k XOUT series, limits crystal drive

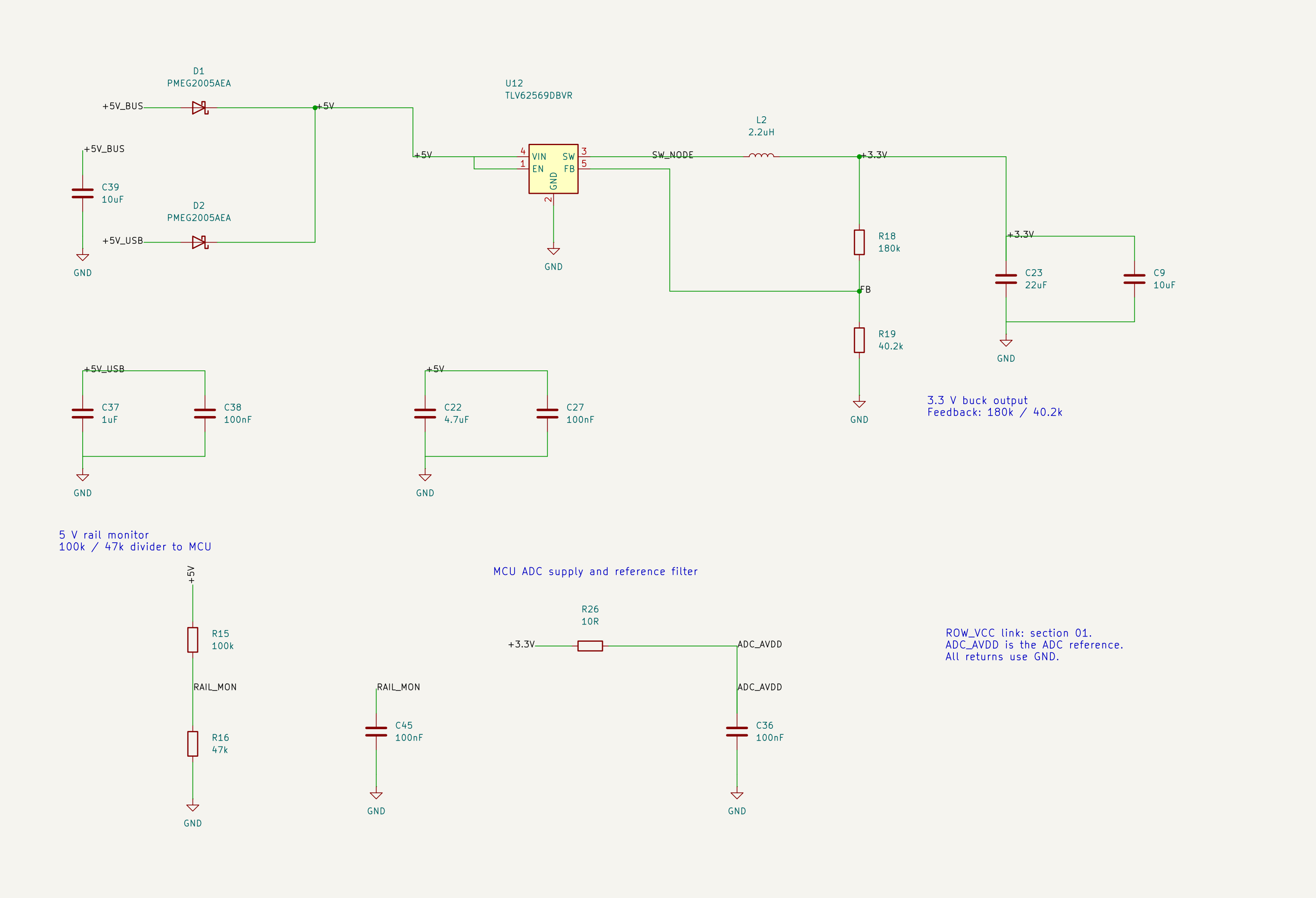
Core regulator / DVDD bypass

MCU 3.3 V bypass

VREG_AVDD: 33R / 4.7uF filter

04 POWER + RAIL MONITOR

Harness or USB power > diode OR > local 3.3 V buck



3.3 V buck output
Feedback: 180k / 40.2k

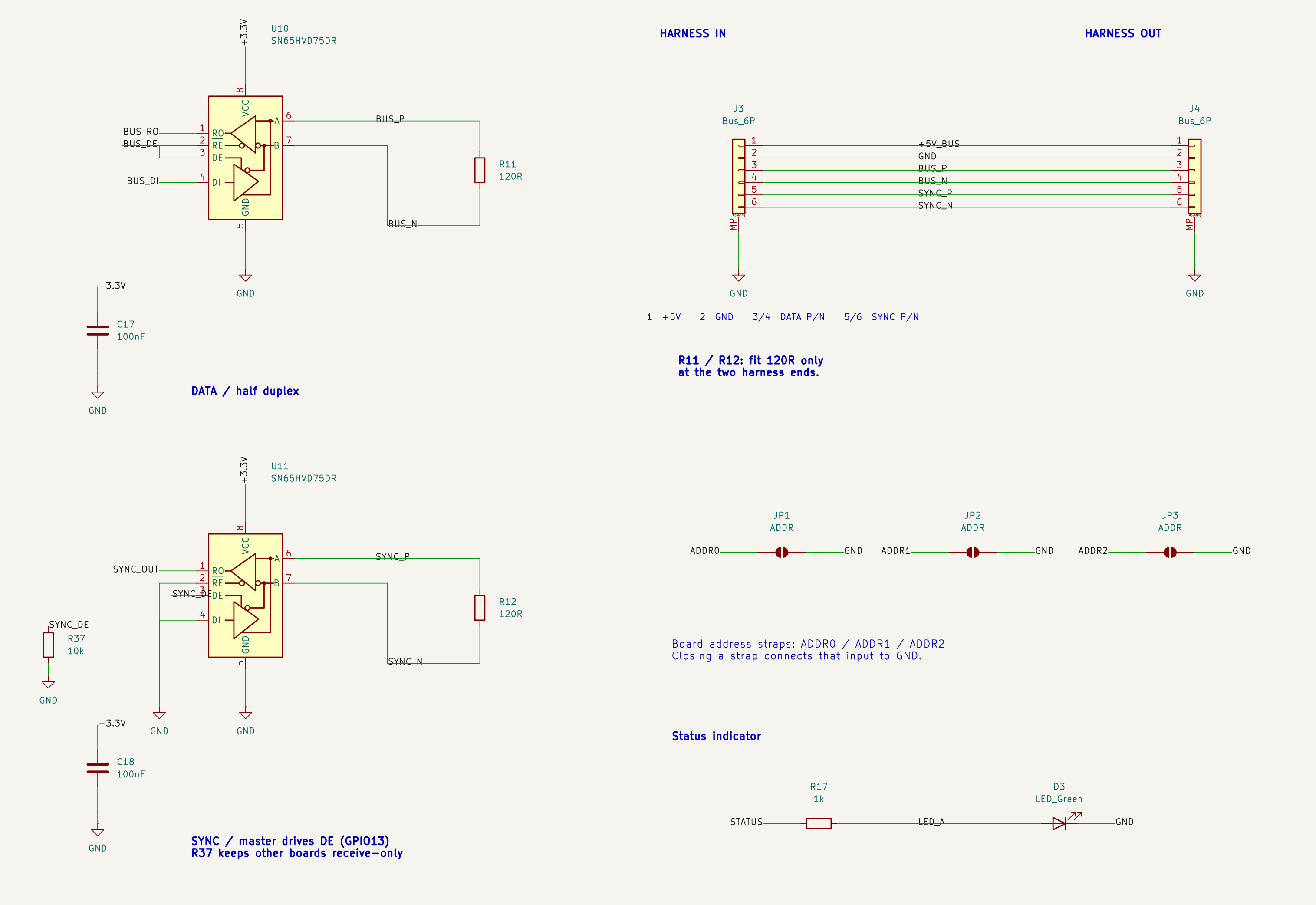
5 V rail monitor
100k / 47k divider to MCU

MCU ADC supply and reference filter

ROW_VCC link: section 01.
ADC_AVDD is the ADC reference.
All returns use GND.

05 COMMUNICATIONS + BOARD ADDRESS

Half-duplex data pair and dedicated receive-only sync pair



DATA / half duplex

1 +5V 2 GND 3/4 DATA P/N 5/6 SYNC P/N

R11 / R12: fit 120R only
at the two harness ends.

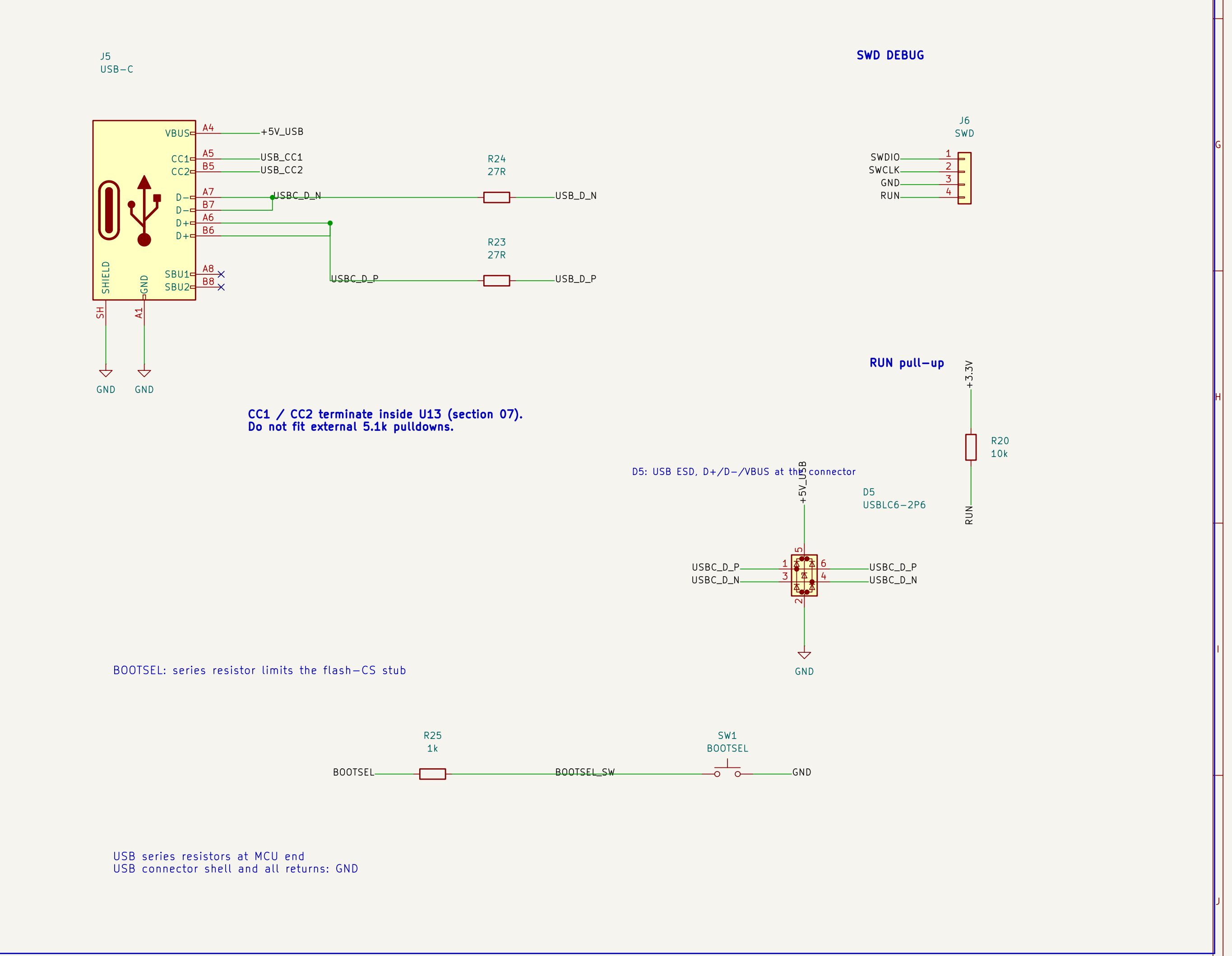
Board address straps: ADDR0 / ADDR1 / ADDR2
Closing a strap connects that input to GND.

Status indicator

SYNC / master drives DE (GPIO13)
R37 keeps other boards receive-only

06 USB + PROGRAMMING

USB-C data / configuration straps / SWD / BOOTSEL



CC1 / CC2 terminate inside U13 (section 07).
Do not fit external 5.1k pulldowns.

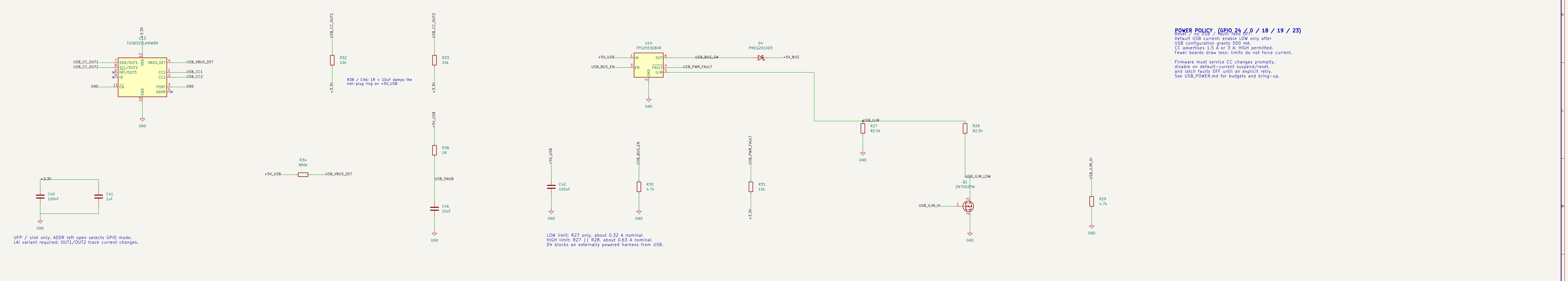
D5: USB ESD, D+/D−/VBUS at the connector

BOOTSEL: series resistor limits the flash-CS stub

USB series resistors at MCU end
USB connector shell and all returns: GND

07 USB-C POWER TO THE HARNESS

Computer port current detection > firmware authorization > current limit > reverse barrier > +5V_BUS



R38 / C46: 1R + 10uF damps the
hot-plug ring on +5V_USB

UFP / sink only; ADDR left open selects GPIO mode.
LAI variant required: OUT1/OUT2 track current changes.

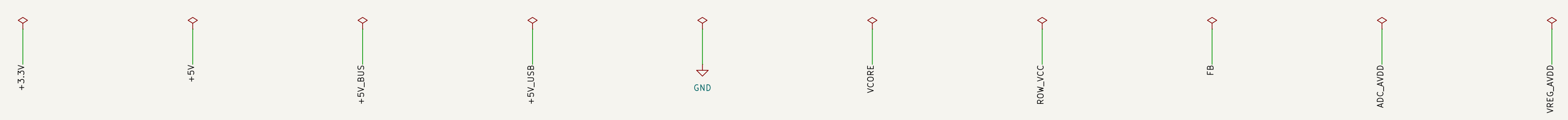
LOW limit: R27 only, about 0.32 A nominal
HIGH limit: R27 || R28, about 0.63 A nominal
D4 blocks an externally powered harness from USB.

POWER POLICY (GPIO 24 / 0 / 18 / 19 / 23)

Reset / no USB / fault: feed OFF.
Default USB current: enable LOW only after
USB configuration grants 500 mA.
CC advertises 1.5 A or 3 A: HIGH permitted.
Fewer boards draw less; limits do not force current.

Firmware must service CC changes promptly,
disable on default-current suspend/reset,
and latch faults OFF until an explicit retry.
See USB_POWER.md for budgets and bring-up.

POWER-NET DRIVE DECLARATIONS



Reading the sheet

Matching net labels connect between sections. A dot marks a branch; an X marks an unused pin.
Bypass capacitors are grouped by supply. USB attach current and the complete eight-board
startup / operating budget require measurements; schematic checks cannot establish them.

Sensor path: ROW_n > mat element > COL_n > SENSE_A/B > AMP_A/B > ADC_A/B

R1/R2 and gain-network values remain fit-on-test values from the existing design.

3.3 V logic | USB-C data and protected chain power | SWD debug
Row drive / dual-bank sensing / RP2354A / RS-485

Sheet: /
File: rev4.kicad_sch

Title: TaxelScan | 32 x 32 sensor interface

Size: User
Date:
Rev: 3
KiCad E.D.A. 10.0.6
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